MILITARY SPECIFICATION

CAP, SPIKE, FOR CARTRIDGE, 105MM, HEAT-T, M456A1E1

1. SCOPE

1.1 This specification covers the spike cap for use with Cartridge, 105MM, HEAT-T, M456A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons; General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes.

DRAWINGS

ARMY

9271036	- Cap, Spike.

FSC: 1315

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Applicable for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 9271036, as applicable, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be of the proper finish and free of cracks, burrs, chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.4 First Article Inspection.-This specification provides for submission of First Article Sample inspection in paragraph 4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of Product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to initiation of regular production, the contractor shall submit a first article sample to the Government facility designated by the contracting officer for evaluation in accordance with the provisions of paragraph 4.2.2. The first article sample shall consist of 25 spike caps which have been produced by the contractor, or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted

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unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government).

4.2.2 Inspection to be Performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification, and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any part fails to comply with any of the applicable requirements for the inspections to be performed in accordance with 4.3.2 and 4.3.3, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly or component in the sample to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods in accordnace with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspections lots (see 4.3.1) of the spike caps designated in the applicable

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paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to the individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cap, Spike (See dwg. 9271036).

Categories	Defects	Method of Inspection (see 6.2)
Critical:	None defined.	
Major:	AQL 1.50 percent	
101.	Flange diameter incorrect	Gage
102.	Runout of flange diameter incorrect	Gage
103.	Lucky cavity diameter incorrect	Gage
104.	Runout of lucky cavity diameter incorrect	Gage
105.	Distance from edge of flange to bottom of lucky cavity incorrect	Gage
106.	Groove diameter incorrect	Gage
107.	Runout of groove diameter incorrect	Gage
108.	Distance from edge of flange to far face of groove incorrect	Gage
109.	Overall length incorrect	Gage
110.	Body diameter incorrect	Gage
111.	Runout of body diameter incorrect	Gage
112.	Body frontal radius over max	Gage
113.	Thread pitch diameter incorrect	Gage

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Minor: AQL 2.50 percent

201.	Flange thickness incorrect	Gage
202.	Minor diameter of threads over max	Gage
203.	Groove inside radius incorrect	Gage
204.	Lucky cavity inside radius incorrect	Gage
205.	Flange frontal radius over max	Gage
206.	Evidence of poor workmanship (see 3.3)	Visual

4.3.3 Testing

Hardness - Major Defect - (See dwg. 9271036).-One of the two following sampling plans shall be used, depending upon the type stock or the treatment process used.

a. Furnace Heat Treatment.-Hardness tests shall be performed on fifteen (15) sample spike caps from each heat treat batch. If any sample fails to comply with the applicable requirements, the batch shall be rejected.

b. Pre-Hardened Stock.-Hardness test samples shall be selected from each lot of spike caps using the sampling plan specified in MIL-STD-105, Inspection Level II, with an AQL of 1.0 percent (%) to determine compliance with the applicable requirement.

Testing shall be as specified in 4.4, using Contractor Designed equipment in accordance with 4.3.4.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.3.2 and in 4.3.3 (See 6.2).

4.4 Test Methods and Procedures

Hardness.-The hardness test shall be performed near the periphery of the front face of the spike cap. The test shall be made using the procedure for performing this test as prescribed by ASTM-E-18.

5. PREPARATION FOR DELIVERY

Packing and Marking

Level C.-Packing and marking shall be in accordance with Standard MIL-STD-1169.

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6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include

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appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect numbers from Section 4 of this specification shall be referenced on each contractor design, together with that component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 It is noted that the size of inspection lots of component assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitiate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

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6.5 Submission of Test Data and Ammunition Data Cards. - In addition to the normal distribution of records, when the assembly is procured by the Army Armament Command, one copy of all ammunition data cards shall be forwarded to each of the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-A and SARPA-AD-E-A, Dover, New Jersey, 07801.

Custodian:
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Preparing Activity:
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